Automated Road Network Segmentation from Satellite Images Using a Modified U-Net Model

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***Abstract*—Extraction of road networks from high-resolution satellite images plays a vital role in many applications such as urban planning, intelligent navigation, disaster response, and Geographic Information Systems (GIS). However, accurate extraction of road networks from HR satellite images remains a challenging task due to complex features such as shadow, vegetation, buildings, and varying lighting conditions, etc., present in satellite images, which make it difficult to distinguish road regions from non-road regions. Traditional image processing approaches using intensity values are not effective and often result in noisy and fragmented output, which is not suitable for large-scale applications.**

In this research, a structured approach has been adopted, transitioning from a traditional image processing approach to a deep learning approach. First, a traditional image processing approach has been implemented, establishing a baseline using grayscale transformation, median filtering, Contrast Limited Adaptive Histogram Equalization, and Otsu thresholding. Then, a simplified version of the U-Net model was implemented, and finally, a modified U-Net model was proposed, in which each encoder, bottleneck, and decoder consists of two convolutional layers along with batch normalization.

The models were evaluated using the Massachusetts Roads Dataset, which contains 1,171 images of 256x256 pixels each. The Modified U Net model outperformed the simplified U Net, improving pixel accuracy from 96.0 percent to 97.1 percent, Dice score from 0.60 to 0.72, and Intersection over Union from 0.43 to 0.56. The proposed model also produced less fragmented outputs.

Index Terms: Satellite image segmentation, road network extraction, U Net, batch normalization, deep learning, remote sensing, semantic segmentation

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I. Introduction

Road infrastructure serves as the foundation of modern society, affecting transportation of people, movement of goods, and the effectiveness of emergency services during crucial times. Keeping precise and current road maps is crucial for urban planning, logistics management, and emergency response. Satellite imagery has emerged as a key source of data for large-scale road mapping because of its extensive spatial coverage and improved resolution, offering an alternative to the high costs and logistical challenges of ground surveys [9].

Historically, extracting roads from remote sensing images depended on manual mapping, a process that was labor-intensive, costly, and susceptible to mistakes. Processing efficiency was later enhanced by rule-based automation methods, though they brought about new constraints. These methods rely on manually set thresholds and morphological filters that need to be adjusted again for various sensors, environmental conditions, and geographic areas.

The rapid expansion of high-resolution imagery, combined with progress in deep learning, has led to convolutional neural networks (CNNs) becoming a strong alternative for road extraction tasks. These models can learn complex and scene-specific features directly from labeled data, thereby reducing the need for manual feature engineering [8],[40].

Among encoderdecoder structures intended for dense pixel-wise prediction, U-Net [2] has shown robust performance. Its encoder-decoder architecture, along with skip connections, allows the model to capture both global context and detailed spatial information. However, the standard U-Net architecture, which employs single convolution layers per block, might have difficulty identifying subtle road features such as thin roads under vegetation, shadowed surfaces, or partially occluded intersections. These constraints emphasize the importance of enhancing the architecture.

This study presents three main contributions. First, it provides a detailed comparison of three road extraction methods: Otsu thresholding [1] as a baseline technique, a simplified U-Net [2] as an existing deep learning method, and a Modified U-Net as the proposed approach. Second, it incorporates dual-convolution blocks with Batch Normalization at every stage of the network, leading to better performance in terms of the Dice coefficient and Intersection over Union (IoU) without notably raising computational costs. Third, it creates a repeatable experimental framework using the Massachusetts Roads Dataset [10] and assesses performance through pixel accuracy, Dice score, and IoU metrics.

The remainder of the paper is structured as follows. Section II discusses relevant work. Section III outlines the dataset and the preprocessing procedures. Section IV outlines the methods that were employed. Section V presents the experimental results, while Section VI concludes with recommendations for future research, focusing on pixel accuracy, Dice score, and IoU metrics.

II. Related Work

Road extraction from aerial and satellite imagery has a long research history spanning classical image processing and modern deep learning.

The problem of road extraction from remote sensing imagery has been studied for several decades, and the literature can be broadly organized into three major phases: classical image processing methods, early deep learning approaches, and modern architecture-driven techniques. This section reviews the most significant contributions across each of these phases.

A. Classical and Threshold-Based Methods

The earliest road extraction methods utilized hand-crafted features and traditional image processing methods. Otsu [1] provided a global thresholding approach, which is extremely illumination dependent and tends to yield a lot of fragmented output[cite: 95, 1109]. Mayer et al. [2] proposed a multi-cue framework that combined a geometric feature and a contextual feature, but this approach requires a lot of parameter tuning and does not generalize well[cite: 99]. Steger [3] created a structure detector that is designed to find road-like patterns, but struggles with heavy occlusions and shadows.

Wegner et al. [4] combined Conditional Random Fields (CRFs) with other methods to enhance spatial consistency, though high processing demands limit scalability. The limitations of these manual and rule-based systems necessitated the shift toward deep learning systems for road extraction[cite: 121].

B. Fully Convolutional Networks for Semantic Segmentation

Long et al. [5] introduced Fully Convolutional Networks (FCNs), enabling end-to-end pixel-wise semantic segmentation. Noh et al. [6] extended this with an encoder–decoder architecture for finer-scale predictions.

Badrinarayanan et al. [7] proposed SegNet, which improves decoding efficiency by reusing pooling indices, making it suitable for real-world applications.

Simonyan and Zisserman [8] developed VGGNet, providing strong feature representations that remain widely used as backbones in segmentation models.

C. U-Net and Its Variants

Ronneberger et al. [9] developed U-Net, an encoder-decoder architecture with skip connections that combine spatial and semantic features for improved accuracy[cite: 104, 1111]. It is highly effective in remote sensing due to its performance even with limited training data.

Oktay et al. [10] enhanced U-Net by incorporating attention mechanisms to focus on pertinent areas. Zhou et al. [11] proposed UNet++, which improves feature fusion by reconfiguring skip connections. Liu et al. [12] presented DeepUNet, which integrates residual connections to facilitate deeper architectures and enhance feature learning.

D. Optimization, Normalization, and Dense Connectivity

Progress in optimization and regularization has significantly impacted the performance of deep road extraction models. The Adam optimizer [13] is particularly beneficial for road segmentation tasks, assisting in faster convergence[cite: 506, 1117].

Batch Normalization (BN) [14] facilitates faster convergence and higher learning rates by stabilizing the training process[cite: 30, 1118]. For cases involving smaller batch sizes, Layer Normalization [15] offers normalization over feature channels.

When reusing features, DenseNet [16] stands out by connecting each layer to all preceding ones. Similarly, shortcut connections in Residual Networks (ResNet) [17] allow for efficient training of very deep models; pre-trained ResNet encoders are frequently used for road segmentation.

E. Multi-Scale Learning and Attention Mechanisms

Road extraction requires multi-scale feature representation to handle varying road widths and resolutions. DeepLabv3+ [18] accomplishes this through Atrous Spatial Pyramid Pooling (ASPP)[cite: 1119].

The Dual Attention Network (DANet) [19] employs channel and spatial self-attention to acquire global context. Spatial CNN (SCNN) [20] enhances the modeling of continuous structures, such as roads, by facilitating spatial information flow.

The EfficientNet model [21] achieves a superior balance between accuracy and efficiency through a compound scaling method that modifies model depth, width, and resolution.

F. Domain-Specific Road Extraction from Remote Sensing Imagery

Mnih and Hinton's early work [22] demonstrated that convolutional networks work well for aerial road detection[cite: 95]. Zhang et al. [23] proposed a residual U-Net to improve boundary visibility and reduce false positives.

Multi-task learning methods [24] predict road masks and centerlines simultaneously. BRRNet [25] employs dilated residual blocks to enlarge receptive fields while maintaining resolution. Contrastive learning [26] enhances accuracy through self-supervised pre-training.

Topology-aware loss functions [27] improve road connectivity by penalizing structural errors like broken segments. Finally, Vision Transformers (ViTs) [28] and architectures like Swin Transformer [29] facilitate global context modeling via hierarchical structures.

III. Dataset and Preprocessing

A. Dataset

Massachusetts Roads Dataset [22] was used for all our experiments. This is a benchmark for aerial road segmentation that can be found on Kaggle platform. The Massachusetts Roads Dataset has high-resolution TIFF images and binary ground-truth masks. In these masks the Massachusetts Roads Dataset marks road pixels as white and background pixels as black. The features related to Massachusetts Roads Dataset are listed in Table I.

B. Preprocessing

Raw 1500×1500-pixel tiles are too big for mini-batch training.Each image is resized to 256×256 pixels. This is done using bilinear interpolation.All pixel values are changed to be between 0 and 1.This helps reduce covarient shift and

TABLE I
MASSACHUSETTS ROADS DATASET CHARACTERISTICS

Parameter	Value / Description
Source	Massachusetts Roads Dataset (Kaggle)
Raw resolution	$1500 \times 1500 \times 3$ (TIFF)
Processed res.	$256 \times 256 \times 3$ (PNG)
Total images	1,171
Training set	1,108 images
Validation set	14 images
Test set	49 images
Colour space	RGB satellite → binary road mask

aceelerate convergence in the data and makes the model learn faster.The mask images are changed so that road pixels are 1 and non-road pixels are 0.No extra transformations are added to the images, in this study.This could be a way to improve the model in the future.

IV. METHODOLOGY

Evaluation of road extraction is performed using three progressively more capable methods.

A. Baseline: Otsu Thresholding

The baseline pipeline applies purely classical operations in sequence (Fig. 1). An input RGB tile is first converted to a single-channel grayscale representation:

$$I_g = 0.299\,R + 0.587\,G + 0.114\,B. \tag{1}$$

A 3×3 median filter suppresses salt-and-pepper noise while preserving edge sharpness. CLAHE then redistributes the local intensity histogram in overlapping sub-windows, clipping the gradient to prevent noise amplification.

Segmentation proceeds via Otsu's criterion [1]. Given the image histogram $p(i)$, the algorithm identifies the threshold t^* that maximises the between-class variance:

$$\sigma_b^2(t) = \omega_0(t)\,\omega_1(t)\left[\mu_0(t) - \mu_1(t)\right]^2, \tag{2}$$

where ω_k and μ_k are the cumulative probability and mean intensity of class $k \in \{0, 1\}$. Pixels with intensity $\geq t^*$ are labelled road (white); the remainder are background (black). Morphological closing and small-region removal are applied as post-processing to improve road continuity.

Input Image → Resize → Grayscale → Median Filter
→ CLAHE → Otsu Threshold → Binary Mask
→ Post-Processing → Road Extraction Output

Fig. 1. Block diagram of the Otsu-thresholding baseline pipeline.

Despite its simplicity and low computational cost, the baseline suffers from well-documented limitations. Buildings, concrete pavements, and sandy ground share intensity values with asphalt, generating substantial false-positive detections. Shadowed road stretches are systematically assigned to the dark class. Since each pixel is evaluated independently, the method cannot exploit the elongated, connected geometry that distinguishes roads from scattered bright objects.

B. Existing System: Simplified U-Net

To move beyond pixel-intensity reasoning, we implement a simplified U-Net [9] that assigns a class label (road or non-road) to every pixel jointly.

1) Architecture Overview

The network follows the canonical encoder–bottleneck–decoder topology. Each encoder block contains a single 3×3 convolution, ReLU activation, and 2×2 max-pooling:

$$y(i, j) = \textstyle\sum_m \sum_n x(i{+}m, j{+}n) \cdot w(m, n), \tag{3}$$

$$\mathrm{ReLU}(x) = \max(0, x), \tag{4}$$

$$y_{\mathrm{pool}}(i, j) = \max_{(m,n)\in W} x(i{+}m, j{+}n). \tag{5}$$

The bottleneck applies a single convolution with 128 filters. The decoder upsamples spatially, concatenates the corresponding encoder feature map via skip connections, and restores the original resolution. A 1×1 convolution followed by sigmoid activation produces a per-pixel road-probability map:

$$\hat{y} = \sigma(w \cdot x + b) = \frac{1}{1 + e^{-(w \cdot x + b)}}. \tag{6}$$

A threshold of 0.5 converts this map into a binary segmentation mask.

2) Training Configuration

The model is optimised with Binary Cross-Entropy (BCE) loss and the Adam optimiser [13] at a learning rate of 10^{-3}, batch size 16, over 50 epochs. Table II lists the full configuration.

TABLE II
TRAINING HYPERPARAMETERS

Parameter	Value
Optimiser	Adam
Learning rate	0.001
Loss function	Binary Cross-Entropy
Batch size	16
Epochs	50
Validation imgs	14

While the simplified U-Net markedly outperforms the Otsu baseline, limiting each block to a single convolution constrains feature-representation depth. The absence of normalisation causes feature-distribution drift across layers, slowing convergence and occasionally producing unstable training curves.

C. Proposed System: Modified U-Net

The Modified U-Net addresses the shortcomings of the simplified version through two key changes: (i) dual convolutional layers per block and (ii) Batch Normalization after each convolution.

1) Dual Convolutional Layers

Replacing single convolutions with stacked pairs allows the network to model richer, more hierarchical feature representations. The first convolution captures low-level patterns (edges, texture gradients), while the second integrates these into higher-order structures (lane markings, road boundaries).

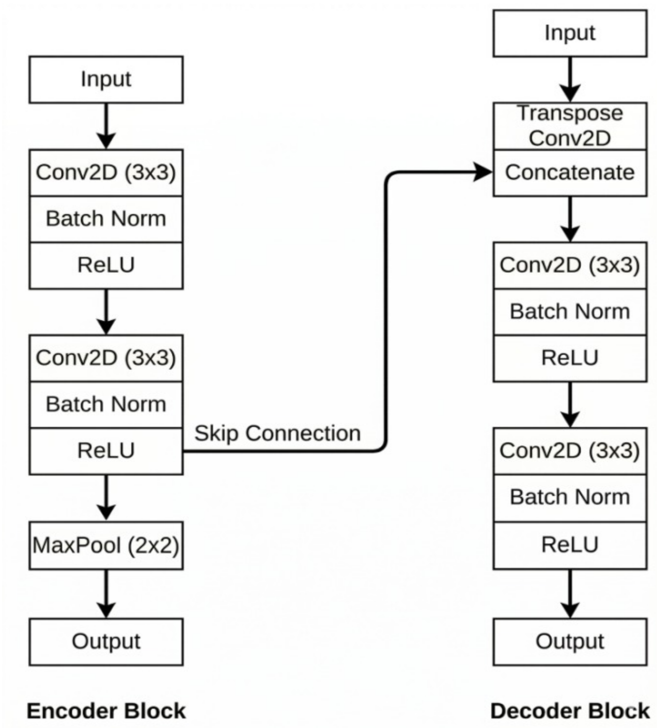

Fig. 2. Proposed System Methodology Overview.

This increased depth is particularly beneficial for detecting thin roads and resolving ambiguities at road–shadow boundaries. Table III provides a component-level comparison between the two deep-learning systems.

TABLE III
LAYER-OPERATION COMPARISON: EXISTING VS. PROPOSED

Block	Simple U-Net	Modified U-Net
Encoder	Conv→ReLU→MaxPool	Conv→BN→ReLU→ Conv→BN→ReLU
Bottleneck	Conv→ReLU	Conv→BN→ReLU→ Conv→BN→ReLU
Decoder	Up→Conv→ReLU	Up→Concat→Conv→ BN→ReLU→Conv→ BN→ReLU

2) Batch Normalization

Batch Normalization [14] normalises the output of each convolutional layer over the current mini-batch before the non-linearity. As detailed in Table IV, this stabilization allows for faster convergence and improved generalization by reducing internal covariate shift. For a mini-batch $\mathcal{B} = \{x_1, \ldots, x_n\}$, the transformation follows the standard normalization and affine rescaling steps:

Step 1 — Batch mean:

$$\mu_{\mathcal{B}} = \frac{1}{n}\sum_{i=1}^{n} x_i. \tag{7}$$

Step 2 — Batch variance:

$$\sigma_{\mathcal{B}}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu_{\mathcal{B}})^2. \tag{8}$$

Step 3 — Normalise:

$$\hat{x}_i = \frac{x_i - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}}. \tag{9}$$

Step 4 — Affine rescaling:

$$y_i = \gamma\,\hat{x}_i + \beta, \tag{10}$$

TABLE IV
IMPACT OF BATCH NORMALIZATION ON THE PROPOSED SYSTEM

Aspect	Effect
Training speed	Normalised activations allow larger effective learning rates, reducing epoch count to convergence.
Stability	Internal covariate shift is reduced, yielding smoother loss curves.
Accuracy	Stabilised training translates to improved final segmentation metrics.
Generalisation	Acts as an implicit regulariser, lowering the risk of over-fitting.

3) Encoder and Decoder Block Architecture

The Modified U-Net's structural improvements, shown in Fig. 2, make it a stronger way to extract features than the standard model. Each Encoder Block has a dual-layer strategy that uses two 3×3 convolutions. Batch Normalization is very important because we do it right after every convolution and before the ReLU activation. This particular order is essential for keeping the activation distributions stable, which stops the "drift" that can happen inside a deep network and slow down training. After the second ReLU, a 2×2 MaxPooling layer makes the spatial dimensions smaller, which helps the model understand the satellite's larger, global context.

The process starts in the Decoder Block by using a skip connection to combine the upsampled feature map with the high-resolution map from the Encoder Block. This combined map then goes through another dual-convolutional sequence to make the spatial details that were recovered during upsampling even better. The blocks work together to make sure that road boundaries stay sharp and continuous by balancing the encoder's ability to capture the "global picture" with the decoder's ability to focus on fine-grained detail. The system gets a high pixel accuracy of 97.1% because the layers work together in this way.

4) End-to-End Workflow

Fig. 3 illustrates the complete inference pipeline. A 256×256 RGB tile is fed to the Modified U-Net, which outputs a per-pixel probability map. Thresholding at 0.5 produces an initial binary mask. Morphological closing fills gaps within detected roads, while morphological opening removes isolated noise regions too small to constitute real road segments. Performance is evaluated against the ground-truth mask using the metrics defined below.

Input Image → Pre-Processing → Modified U-Net (Encoder–Bottleneck–Decoder) → Probability Map → Thresholding → Post-Processing → Output → Evaluation (Accuracy, IoU, Dice)

Fig. 3. End-to-end workflow of the proposed Modified U-Net system.

D. Evaluation Metrics

Three complementary metrics assess segmentation quality at the pixel level. Let TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives respectively.

Pixel Accuracy:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \tag{11}$$

Dice Coefficient (F1 score):

$$\text{Dice} = \frac{2\,|P \cap G|}{|P| + |G|} = \frac{2TP}{2TP + FP + FN}. \tag{12}$$

Intersection over Union (IoU / Jaccard Index):

$$\text{IoU} = \frac{TP}{TP + FP + FN}. \tag{13}$$

Accuracy alone can be misleading when road pixels are a small fraction of the image. Dice and IoU focus on the foreground class and are therefore more informative for imbalanced segmentation tasks.

V. Results

A. Otsu Thresholding (Baseline)

Results from classical pipelines serve as a useful baseline for comparison. The shortcomings of intensity-based segmentation can be seen in the resulting binary mask, where segmented road regions are more fragmented than integrated. The segmentation also breaks apart due to differences in the appearance of the road caused by shadows, uneven surfaces, and changes in lighting.

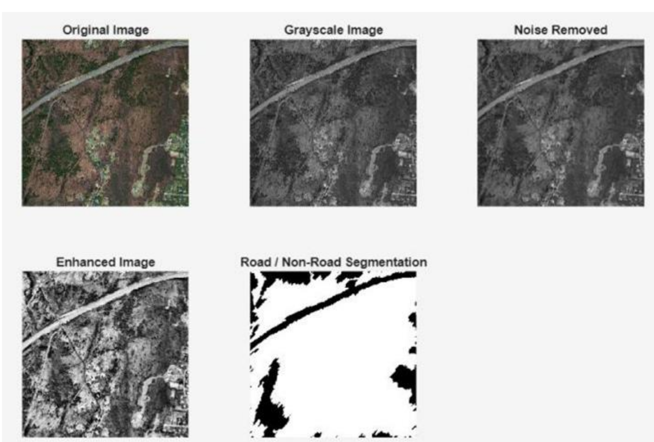

Fig. 4. Otsu Thresholding Outputs: Original, Grayscale, Noise Removed, Enhanced, and Final Result.

The Fig. 4 illustrates several false positive segmentation results. In the bright regions of the image, rooftops, light sand, and painted roads are incorrectly classified as roads due to the similar intensity values of these regions to those of real road surfaces. The misclassifications appear as disconnected white spots in the output mask and do not connect to the actual road network. On the other hand, road segments that are darker, especially if they are in the shadow, are not included in the segmentation output. Because of this, there are gaps in the binary mask which result in false negatives.

B. Simplified U-Net (Existing)

To overcome the limitations of classical methods, a simplified U-Net model serves as a baseline in deep learning. As shown in Fig. 5, this model greatly improves segmentation quality by learning spatial and contextual features directly from data [5], [9].

The predicted outputs show a clear enhancement in capturing the overall structure of the road network. Unlike the classical approach, the U-Net model can identify continuous road segments across many regions of the image. The encoder-decoder architecture with skip connections helps the model keep both high-level semantic information and low-level spatial details [7], [9].

However, some limitations persist. The segmentation output still shows some discontinuities, especially in narrow roads and complex intersections. Some regions have incomplete or broken road segments, indicating that the model has difficulty capturing fine details.

Additionally, minor false positives appear, where non-road areas are mistakenly labeled as roads. This suggests that while the simplified U-Net effectively captures general patterns, it lacks the depth and feature refinement necessary for highly accurate segmentation. Similar issues have been noted in standard segmentation networks [11], [18].

Therefore, although the simplified U-Net marks a significant step forward compared to classical methods, there is still room for improvement in accuracy and structural consistency.

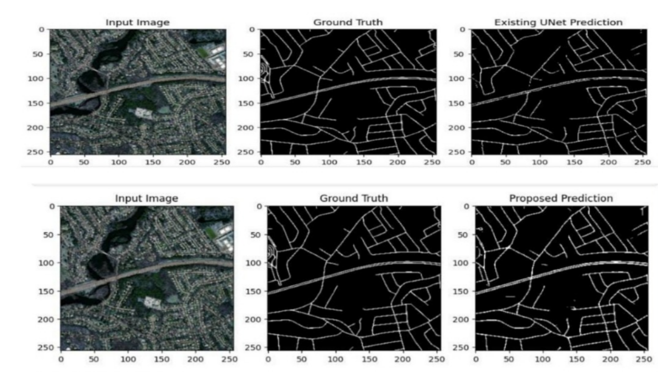

Fig. 5. Simplified U-Net (Existing System) Prediction Results.

C. Modified U-Net (Proposed)

The proposed Modified U-Net architecture introduces enhancements to address the limitations observed in the simplified model. As illustrated in Fig. 6, the segmentation results obtained using the proposed approach show a marked improvement in both accuracy and visual quality.

The predicted road masks are significantly more continuous and closely aligned with the ground truth. The model effectively captures both major roads and finer road structures, maintaining connectivity even in challenging regions such as intersections and shadowed areas. This behavior is consistent with the strengths of encoder–decoder architectures for semantic segmentation [5], [9].

The inclusion of dual convolutional layers in each block enhances the model's ability to extract detailed features at multiple levels. In addition, Batch Normalization [14] stabilizes the training process and improves generalization, resulting in more consistent predictions across different images. Similar improvements have been observed in deep architectures such as ResNet and DenseNet [16], [17].

Compared to the existing system, the proposed model produces fewer false positives and significantly reduces noise in the segmentation output. Road boundaries are more clearly defined, and thin road segments are better preserved. This improvement in structural consistency aligns with recent advancements in segmentation models [11], [18].

These improvements demonstrate that the architectural modifications lead to stronger feature representation and better learning of complex spatial patterns. As a result, the Modified U-Net provides a more reliable and accurate solution for road extraction from satellite imagery.

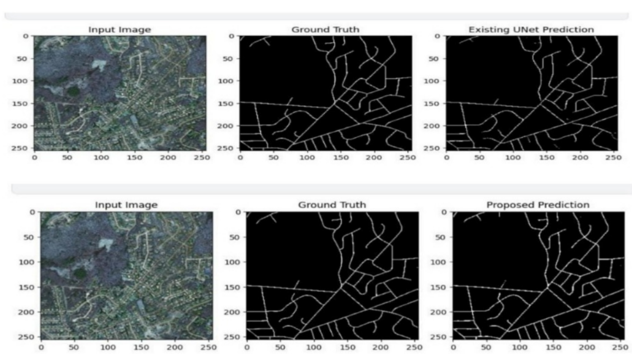

Fig. 6. Modified U-Net (Proposed System) Prediction Results.

D. Graphical Analysis

The graphical representation of the results provides a clearer understanding of the performance differences between the existing and proposed models. As shown in Fig. 7, the relationship between model accuracy and computational complexity is illustrated. The simplified U-Net [9] achieves moderate accuracy with lower complexity, while the Modified U-Net demonstrates higher accuracy with a slight increase in computational cost. This indicates that the proposed model effectively balances performance improvement with computational efficiency, which is consistent with findings in deep convolutional architectures [17], [21].

Fig. 8 presents a bar graph comparing key performance metrics, including pixel accuracy, Dice coefficient, and Intersection over Union (IoU), for both models. The proposed Modified U-Net consistently outperforms the simplified U-Net across all metrics. The improvement in Dice coefficient and IoU is particularly significant, as these metrics are widely used for evaluating segmentation performance [9], [18].

The graphical results clearly support the quantitative findings presented in the tables. The proposed model not only improves numerical performance but also demonstrates more stable and consistent behavior across evaluation metrics. Overall, the graphs provide strong visual evidence of the effectiveness of the Modified U-Net for road extraction tasks, aligning with recent advances in deep learning-based segmentation methods [5], [11].

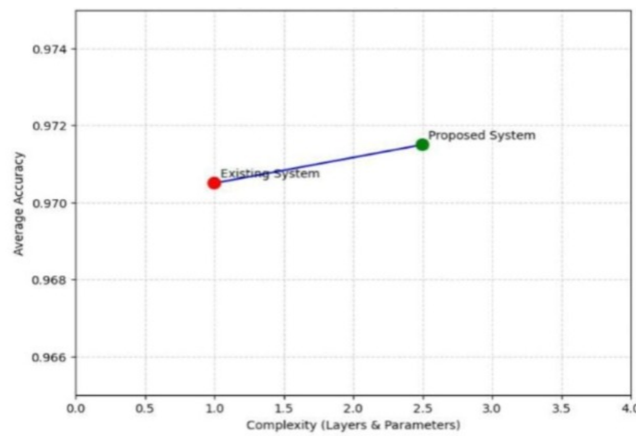

Fig. 7. Comparison Graph: Accuracy vs. Computational Complexity

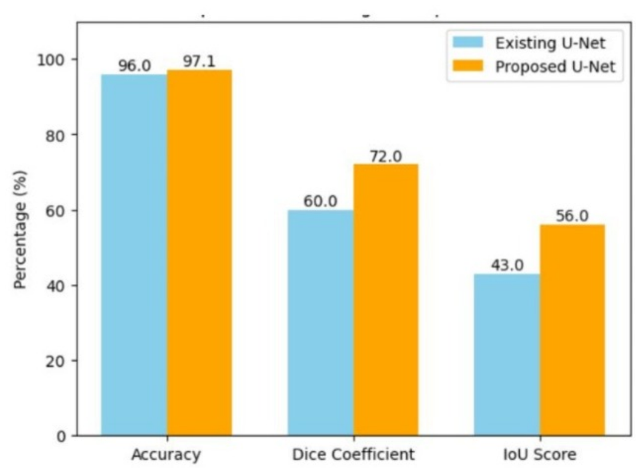

Fig. 8. Comparison of existing and proposed u-net

E. Quantitative Comparison

Table V reports pixel accuracy, Dice coefficient, and IoU for the two deep-learning models on the 49-image test set.

TABLE V
QUANTITATIVE PERFORMANCE COMPARISON

Model	Acc. (%)	Dice	IoU
Simple U-Net (existing)	96.0	0.60	0.43
Modified U-Net (proposed)	97.1	0.72	0.56

The proposed model achieves a 1.1-percentage-point gain in accuracy, a 20% relative improvement in Dice (0.60→0.72), and a 30% relative improvement in IoU (0.43→0.56). These gains are consistent across the test split, suggesting genuine generalisation benefit rather than over-fitting to the training data.

Table VI summarises a qualitative comparison across all three systems.

TABLE VI
QUALITATIVE COMPARISON ACROSS ALL THREE SYSTEMS

Property	Otsu	Simplified U-Net	Modified U-Net
Accuracy	Low	Moderate	High
Noise level	High	Moderate	Low
Road continuity	Poor	Moderate	Good
Shadow handling	Poor	Moderate	Better

VI. CONCLUSION AND FUTURE WORK

This study has demonstrated an effective technique for extracting road networks from satellite imagery, combining traditional image processing methods and deep learning methodologies for the extraction process. The initial approach was a baseline technique comprising of an initial step of grayscale conversion, followed by filtering, with the final thresholding using the Otsu method. Although this process was relatively simple to implement, there was poor performance in terms of producing a continuous road networks due to using pixel level intensity to provide roadway geometry; thus, the implementation could not accurately produce a complete roadway when tested on images with complex scenarios.

The next implementation used a U-Net architecture that provided improved results when learning spatial and contextual information from the data. Thus, this architecture used more sophisticated mathematical functions to generate more accurate segmented road networks. However, 'thin' roadways or maintaining roadway connectivity in shadow or occluded areas was still prevalent with this U-Net model.

In order to enhance the overall performance of U-Net, a Modified U-Net was developed. The main change from U-Net to Modified U-Net was the addition of extra convolutional layers and the addition of batch normalization. These alterations have provided enhancements to the overall performance of the network across each of the metrics used to evaluate the model (Accuracy, Dice Score, and Intersection over Union).Moreover, traditional methods, along with the majority of current deep-learning architectures, typically offer road networks that are more continuous and cleaner compared to either enhanced versions of U-Net or newer deep-learning models.

Overall, the outcomes of the work show the effectiveness of deep learning methods, especially the enhanced U-Net, for road extraction tasks, as well as their appropriateness for practical uses such as urban planning, navigation, and disaster management.

Although the study has yielded promising results, there are many opportunities for further improvements. Specifically, there are numerous opportunities to explore advanced architectures like Attention U-Net, UNet++, and transformer-based models that could enhance results. Moreover, employing established augmentation techniques along with larger and more diverse datasets is expected to enhance generalization to various environmental conditions. Additionally, multi-class segmentation of features, like distinguishing between roads, buildings, and vegetation, will further enhance the practicality of this technology.

Future research may focus on enhancing the model's architecture for real-time applications, particularly in collaboration with GIS systems to create a foundation for the eventual deployment of this application.

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